

Self-contact frictional jamming of helical ribbons

Brandon Wong, Ben Foster

August 22, 2026

1 Geometric model

The system can be modelled as a constant-width rectangular ribbon of negligible thickness wrapped around a uniform-radius cylinder at a uniform pitch. One end of the ribbon is fixed in place to prevent rigid body motion, while the other (“active” end) is displaced in the z direction (the axis of the cylinder). Refer to Figure 1 for a visual representation of the system. The system has the following geometric constants:

- L : the length of the unrolled rectangle.
- W : the width of the unrolled rectangle. Assume $L \gg W$.
- θ_L : the total wind angle of the helix, which is fixed based on the boundary conditions (the two ends of the ribbon cannot rotate about the z -axis relative to one another, only translate along it, so the total number of turns is a fixed topological property of how the ribbon was wound and does not change as it is pulled). $\theta_L = 2\pi n$ where n is the number of winds around the z axis (not necessarily integer).

The state of the system is parametrized by the parameter x , which describes how far the active end has displaced in the z direction relative to the base end.

1.1 Coordinate systems

Positions on the helix can be described with local surface u, v coordinates, which describe the unrolled rectangle. The u direction runs along the length of the unrolled rectangle, and the v direction runs across its width. Alternatively, positions can be described in 3d space using cylindrical r, θ, z coordinates.

To simplify the geometry, we assume that the helix has a constant radius r and a constant pitch angle ϕ (the angle the local $\hat{u}$ makes with the r, θ plane). Equivalently, the local orthonormal frame $(\hat{u}, \hat{v})$ tangent to the ribbon surface is related to the cylinder’s own natural frame $(\hat{\theta}, \hat{z})$ by a rotation through the angle ϕ :

$$\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}, \quad \hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}. \quad (1)$$

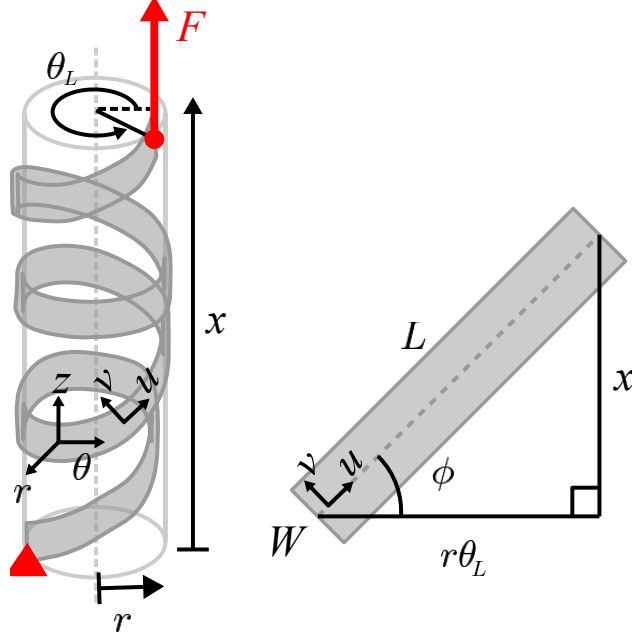


Figure 1: in progress

We now use Eq. (1) to relate the geometric constants L, θ_L to the pitch angle ϕ and radius r at a given extension x . From Eq. (1), $\hat{u}$ has z -component $\sin \phi$, so moving a distance du along the ribbon's own length produces an axial displacement $dz = \sin \phi du$. Since ϕ is constant along the whole ribbon at a given (frozen) extension x , integrating this from the base end ($u = 0$) to the active end ($u = L$) gives a total axial displacement of $L \sin \phi$. By definition, this total axial displacement between the two ends of the ribbon is exactly x , so

$$x = L \sin \phi \quad \implies \quad \phi = \sin^{-1} \left(\frac{x}{L} \right). \quad (2)$$

Likewise, $\hat{u}$ has θ -component $\cos \phi$, so moving a distance du along the ribbon produces a circumferential arc length $r d\theta = \cos \phi du$. Integrating over the whole ribbon length L , the ribbon sweeps out a total circumferential arc length $L \cos \phi$. This total arc length must also equal $r \theta_L$ (arc length equals radius times swept angle), since by construction the ribbon winds through the fixed total angle θ_L regardless of x . Equating the two expressions for this arc length and using $\cos \phi = \sqrt{1 - \sin^2 \phi}$ together with Eq. (2) gives

$$r \theta_L = L \cos \phi \quad \implies \quad r = \frac{L}{\theta_L} \cos \phi = \frac{L}{\theta_L} \sqrt{1 - \frac{x^2}{L^2}} = \frac{\sqrt{L^2 - x^2}}{\theta_L}. \quad (3)$$

1.2 Position vector

An area element $du \times dv$ at point (u, v) has a position vector given in cylindrical coordinates. We build this up from Eq. (1): since $\hat{u} = \cos \phi \hat{\theta} + \sin \phi \hat{z}$, moving along the ribbon by du produces $r d\theta = \cos \phi du$ and $dz = \sin \phi du$; since $\hat{v} = -\sin \phi \hat{\theta} + \cos \phi \hat{z}$, moving across the width by dv produces $r d\theta = -\sin \phi dv$ and $dz = \cos \phi dv$. Integrating these (at fixed, frozen x , so r and ϕ are

constants) from the reference point $(u, v) = (0, 0)$ gives

$$\vec{p}(u, v, x) = \begin{bmatrix} r \\ \theta \\ z \end{bmatrix} = \begin{bmatrix} r \\ \frac{u \cos \phi - v \sin \phi}{r} \\ u \sin \phi + v \cos \phi \end{bmatrix}. \quad (4)$$

1.3 Local curvature

A cylinder of radius r has two principal curvatures: $\kappa_\theta = 1/r$ in the circumferential ($\hat{\theta}$) direction in which it is rolled, and $\kappa_z = 0$ in the axial ($\hat{z}$) direction in which it is completely flat. Equivalently, the standard cylindrical basis vectors rotate with θ at the rates

$$\frac{\partial \hat{r}}{\partial \theta} = \hat{\theta}, \quad \frac{\partial \hat{\theta}}{\partial \theta} = -\hat{r}, \quad \frac{\partial \hat{z}}{\partial \theta} = 0, \quad (5)$$

(the first two follow from writing $\hat{r} = \cos \theta \hat{x} + \sin \theta \hat{y}$ and $\hat{\theta} = -\sin \theta \hat{x} + \cos \theta \hat{y}$ in terms of the fixed Cartesian basis and differentiating; the third holds because $\hat{z}$ does not depend on θ at all — this is exactly the statement $\kappa_z = 0$).

To differentiate the ribbon's own frame $(\hat{u}, \hat{v}, \hat{r})$ with respect to (u, v) at a fixed x (and therefore a fixed r, ϕ), we take the partial derivatives of the θ component of Eq. (4):

$$\frac{\partial \theta}{\partial u} = \frac{\cos \phi}{r}, \quad \frac{\partial \theta}{\partial v} = -\frac{\sin \phi}{r}. \quad (6)$$

Combining Eq. (5) and Eq. (6) via the chain rule (e.g. $\frac{\partial \hat{r}}{\partial u} = \frac{\partial \hat{r}}{\partial \theta} \frac{\partial \theta}{\partial u}$) gives

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos \phi}{r} \hat{\theta}, \quad \frac{\partial \hat{r}}{\partial v} = -\frac{\sin \phi}{r} \hat{\theta}, \quad \frac{\partial \hat{\theta}}{\partial u} = -\frac{\cos \phi}{r} \hat{r}, \quad \frac{\partial \hat{\theta}}{\partial v} = \frac{\sin \phi}{r} \hat{r}, \quad (7)$$

with $\hat{z}$ constant with respect to u, v . Now we take the partial derivatives of $\hat{u}, \hat{v}$ themselves, using Eq. (1) with ϕ held fixed:

$$\begin{aligned} \frac{\partial \hat{u}}{\partial u} &= \cos \phi \frac{\partial \hat{\theta}}{\partial u} = -\frac{\cos^2 \phi}{r} \hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \cos \phi \frac{\partial \hat{\theta}}{\partial v} = \frac{\sin \phi \cos \phi}{r} \hat{r}, \\ \frac{\partial \hat{v}}{\partial u} &= -\sin \phi \frac{\partial \hat{\theta}}{\partial u} = \frac{\sin \phi \cos \phi}{r} \hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\sin \phi \frac{\partial \hat{\theta}}{\partial v} = -\frac{\sin^2 \phi}{r} \hat{r}. \end{aligned} \quad (8)$$

The same three coefficients, $\cos^2 \phi/r$, $\sin^2 \phi/r$, and $\sin \phi \cos \phi/r$ (the last one twice), keep reappearing. We name them for convenience:

$$\kappa_{uu} := \frac{\cos^2 \phi}{r}, \quad \kappa_{vv} := \frac{\sin^2 \phi}{r}, \quad \kappa_{uv} := \frac{\sin \phi \cos \phi}{r}, \quad (9)$$

Finally, we differentiate $\hat{r}$ itself. We already have $\frac{\partial \hat{r}}{\partial u}, \frac{\partial \hat{r}}{\partial v}$ in terms of $\hat{\theta}$ (Eq. 7). To express them in terms of $\hat{u}, \hat{v}$ instead, we invert Eq. (1) to obtain $\hat{\theta} = \cos \phi \hat{u} - \sin \phi \hat{v}$. Substituting this

expression for $\hat{\theta}$ into Eq. (7) gives

$$\frac{\partial \hat{r}}{\partial u} = \frac{\cos \phi}{r} (\cos \phi \hat{u} - \sin \phi \hat{v}) = \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v} \quad (10)$$

$$\frac{\partial \hat{r}}{\partial v} = -\frac{\sin \phi}{r} (\cos \phi \hat{u} - \sin \phi \hat{v}) = -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v} \quad (11)$$

Collecting everything derived in this section,

$$\begin{aligned} \frac{\partial \hat{r}}{\partial u} &= \kappa_{uu} \hat{u} - \kappa_{uv} \hat{v}, & \frac{\partial \hat{r}}{\partial v} &= -\kappa_{uv} \hat{u} + \kappa_{vv} \hat{v}, \\ \frac{\partial \hat{u}}{\partial u} &= -\kappa_{uu} \hat{r}, & \frac{\partial \hat{u}}{\partial v} &= \kappa_{uv} \hat{r}, \\ \frac{\partial \hat{v}}{\partial u} &= \kappa_{uv} \hat{r}, & \frac{\partial \hat{v}}{\partial v} &= -\kappa_{vv} \hat{r}. \end{aligned} \quad (12)$$

2 Stress balance

We will model the paper yoyo as a thin shell of small but finite thickness $t \ll r$. Because the ribbon's thickness ultimately carries the boundary tractions responsible for friction (Section 3), we work throughout with its full three-dimensional stress state. We use $\zeta \in [-t/2, t/2]$ for the coordinate measuring position through the thickness, i.e., displacement from the mid-surface along $\hat{r}$. At a point (u, v, ζ) the material carries the full Cauchy stress tensor:

$$\sigma = \begin{bmatrix} \sigma_{uu} & \sigma_{uv} & \sigma_{ur} \\ \sigma_{uv} & \sigma_{vv} & \sigma_{vr} \\ \sigma_{ur} & \sigma_{vr} & \sigma_{rr} \end{bmatrix} \quad (13)$$

There are six independent components: the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$; the transverse shears σ_{ur}, σ_{vr} by which the ribbon can transmit a load sideways from one part of its width to another, and the normal pressure σ_{rr} caused by layer-layer contact at the ribbon's surface.

We perform a force balance on a differential volume element $du \times dv \times d\zeta$. Each direction has a corresponding traction vector:

$$\vec{t}_u = \sigma_{uu} \hat{u} + \sigma_{uv} \hat{v} + \sigma_{ur} \hat{r} \quad \vec{t}_v = \sigma_{uv} \hat{u} + \sigma_{vv} \hat{v} + \sigma_{vr} \hat{r} \quad \vec{t}_\zeta = \sigma_{ur} \hat{u} + \sigma_{vr} \hat{v} + \sigma_{rr} \hat{r} \quad (14)$$

Because $t \ll r$, the element can be approximated as a rectangular prism, so the net force on the element is

$$\left(\frac{\partial \vec{t}_u}{\partial u} + \frac{\partial \vec{t}_v}{\partial v} + \frac{\partial \vec{t}_\zeta}{\partial \zeta} \right) du dv d\zeta = 0 \quad (15)$$

However, because $\hat{u}, \hat{v}, \hat{r}$ themselves depend on u, v (Eq. 12), differentiating $\vec{t}_u, \vec{t}_v, \vec{t}_\zeta$ produces extra terms beyond the plain derivatives of the stress components. For example, expanding $\frac{\partial}{\partial u}(\sigma_{uu} \hat{u})$ introduces a curvature-weighted term in the $\hat{r}$ direction:

$$\frac{\partial}{\partial u}(\sigma_{uu} \hat{u}) = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} + \sigma_{uu} \frac{\partial \hat{u}}{\partial u} = \frac{\partial \sigma_{uu}}{\partial u} \hat{u} - \kappa_{uu} \sigma_{uu} \hat{r} \quad (16)$$

The additional curvature-weighted terms reflect the fact that the element is not perfectly planar, so in-plane stresses are slightly rotated to have a small out-of-plane component, and vice versa. Expanding all terms and sorting by component gives three scalar force-balance equations:

$$\hat{u} : \boxed{\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu}\sigma_{ur} - \kappa_{uv}\sigma_{vr} = 0} \quad (17)$$

$$\hat{v} : \boxed{\frac{\partial \sigma_{uv}}{\partial u} + \frac{\partial \sigma_{vv}}{\partial v} + \frac{\partial \sigma_{vr}}{\partial \zeta} - \kappa_{uv}\sigma_{ur} + \kappa_{vv}\sigma_{vr} = 0} \quad (18)$$

$$\hat{r} : \boxed{\frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu}\sigma_{uu} + 2\kappa_{uv}\sigma_{uv} - \kappa_{vv}\sigma_{vv} = 0} \quad (19)$$

Equilibrium alone only gives 3 equations for 6 unknown stress components. However, because t is small, we will make a first-order approximation that the stress field varies linearly through t . This means that for any stress component, if we know its boundary value at both $-\frac{t}{2}, \frac{t}{2}$ for all (u, v) , that component is essentially solved. In the following section, we will obtain the surface boundary conditions for σ_{ur}, σ_{vr} , and σ_{rr} based on layer-layer contact.

3 Friction

3.1 Overlapping neighbors

Because the ribbon is wound around itself, a given area element at (u, v) could physically overlap with another area elements one full turn away along the ribbon. Specifically, an area element at point (u, v) overlaps with the neighboring interior layer at point (u^-, v^-) if their cylindrical coordinates have equal r and z , but differ by one full turn in θ . We can express this by taking the difference in $\vec{p}$:

$$\vec{p}(u, v, x) - \vec{p}(u^-, v^-, x) = \begin{bmatrix} 0 \\ 2\pi \\ 0 \end{bmatrix} \quad (20)$$

Plugging in Eq (4) gives:

$$u^- = u - 2\pi r \cos \phi, \quad (21)$$

$$v^- = v + 2\pi r \sin \phi. \quad (22)$$

By the identical argument with $\theta(u^+, v^+) = \theta(u, v) + 2\pi$ (i.e., the neighboring *outer* layer, one full turn further out), the element has an outer overlapping neighbor at

$$u^+ = u + 2\pi r \cos \phi, \quad (23)$$

$$v^+ = v - 2\pi r \sin \phi. \quad (24)$$

Some points (u, v) on the surface overlap with an inner or outer layer, while others don't. We

neglect the edge effects where a neighbor is absent because u^-, u^+ is outside the $[0, L]$ bounds. As a result, an inner neighbor is present exactly whenever v^- lies within $[-W/2, W/2]$, and an outer neighbor is present exactly whenever v^+ lies in $[-W/2, W/2]$. That is,

$$\text{Inner neighbor is present if } v \leq \frac{W}{2} - 2\pi r \sin \phi \quad (25)$$

$$\text{Outer neighbor is present if } v \geq -\frac{W}{2} + 2\pi r \sin \phi \quad (26)$$

3.2 Friction direction

The position vector $\vec{p}(u, v, x)$ of a material point (u, v) is given by Eq. (4). Substituting Eq. (2) and Eq. (3) to expose the dependencies on x gives:

$$\vec{p}(u, v, x) = \begin{bmatrix} \frac{\sqrt{L^2 - x^2}}{\theta_L} \\ \frac{u\theta_L}{L} - \frac{v x \theta_L}{L\sqrt{L^2 - x^2}} \\ \frac{ux}{L} + \frac{v\sqrt{L^2 - x^2}}{L} \end{bmatrix} \quad (27)$$

The velocity of a given material point (u, v) as the active end of the ribbon is displaced (i.e., as x changes) is then obtained by taking the partial derivative of each component of Eq. (4) with respect to x :

$$\frac{\partial}{\partial x} \vec{p}(u, v, x) = \begin{bmatrix} \frac{\partial r}{\partial x} \\ \frac{\partial \theta}{\partial x} \\ \frac{\partial z}{\partial x} \end{bmatrix} = \begin{bmatrix} -\frac{x}{\theta_L \sqrt{L^2 - x^2}} \\ -\frac{L \theta_L v}{(L^2 - x^2)^{3/2}} \\ \frac{u}{L} - \frac{vx}{L\sqrt{L^2 - x^2}} \end{bmatrix}. \quad (28)$$

We can then plug u^-, v^- into Eq. (28) to compute the velocity of the element relative to its inner overlapping neighbor, which gives the relative slip direction in the r, θ, z frame:

$$\frac{\partial \vec{p}(u, v, x)}{\partial x} - \frac{\partial \vec{p}(u^-, v^-, x)}{\partial x} = \begin{bmatrix} 0 \\ \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix}. \quad (29)$$

As expected, there is no relative velocity in the radial direction (i.e., the layers are not lifting away from each other). We can now compute the relative slip direction in the local frame by projecting this slip velocity from the θ, z basis onto the u, v basis using the relation described by Eq. (1). Since $\frac{\partial \theta}{\partial x}$ above is an *angular* rate, we first convert it to a linear (arc-length) rate by multiplying by r , giving $v_\theta = r \frac{\partial \theta}{\partial x}$; then projecting as just described:

$$\vec{v}_{slip}(u, v, x) = \begin{bmatrix} \cos \phi & \sin \phi \\ -\sin \phi & \cos \phi \end{bmatrix} \begin{bmatrix} r \cdot \frac{2\pi x}{L^2 - x^2} \\ \frac{2\pi}{\theta_L} \end{bmatrix} \quad (30)$$

We simplify this expression by substituting x and r in terms of ϕ (Eq. (2), (3)) to expose the double-angle combination, then applying double angle trig identities to factor out

common terms. The result is a scaled unit vector:

$$\vec{v}_{slip}(u, v, x) = \frac{2\pi}{\theta_L \cos \phi} \begin{bmatrix} \sin(2\phi) \\ \cos(2\phi) \end{bmatrix} \quad (31)$$

The prefactor is strictly positive for all $x \in (0, L)$. Therefore, the relative slip velocity points in the direction of the unit vector $-(\sin 2\phi, \cos 2\phi)$ in the local u, v frame, so the Coulomb friction force opposing this motion must point in the opposite direction:

$$\hat{f} = - \begin{bmatrix} \sin(2\phi) \\ \cos(2\phi) \end{bmatrix} \quad (32)$$

3.3 Friction magnitude

Let us take a moment to introduce some shorthand notation:

$$\sigma_{rr}^{\pm} := \sigma_{rr} \left(\zeta = \pm \frac{t}{2} \right) \quad (33)$$

$$\Delta\sigma_{rr} := \sigma_{rr}^+ - \sigma_{rr}^- \quad (34)$$

$$S_{rr} := \sigma_{rr}^+ + \sigma_{rr}^- \quad (35)$$

Recall that Eq. 32 is the friction exerted on the element by the inner layer, which resists the element from sliding forward with x . The components of this friction force are equivalent to the shear stresses acting on the inner ($-\hat{r}$) face. Additionally, the magnitude of the Coulomb friction force is equivalent to the normal pressure $-\sigma_{rr}$ times a friction coefficient μ . By saying this, we assume $\sigma_{rr} \leq 0$ (the element is under compression), deferring a rigorous justification to Section ??.

$$\sigma_{ur} \left(\zeta = -\frac{t}{2} \right) = -\mu \sigma_{rr}^- \sin(2\phi) \quad (36)$$

$$\sigma_{vr} \left(\zeta = -\frac{t}{2} \right) = -\mu \sigma_{rr}^- \cos(2\phi) \quad (37)$$

By the same argument, the friction exerted by the outer layer is the exact opposite, dragging the element forward. We then obtain a similar boundary relationship for the outer ($+\hat{r}$) face. Note that the sign is positive due to the sign convention for shear stresses on the inner face of an element.

$$\sigma_{ur} \left(\zeta = \frac{t}{2} \right) = \mu \sigma_{rr}^+ \sin(2\phi) \quad (38)$$

$$\sigma_{vr} \left(\zeta = \frac{t}{2} \right) = \mu \sigma_{rr}^+ \cos(2\phi) \quad (39)$$

Let thickness-averaged stress components be denoted by a bar, i.e., $\bar{\sigma}_{ij} := \frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \sigma_{ij} d\zeta$. For simplicity, we assume the thickness is small enough that the variation of all stress components through the thickness can be approximated as linear. We can now use Eqs. 36–39 to get the

thickness-averaged values $\bar{\sigma}_{ur}$ and $\bar{\sigma}_{vr}$ in terms of the two boundary values of σ_{rr} :

$$\boxed{\bar{\sigma}_{ur} = -\frac{\mu}{2} S_{rr} \sin(2\phi)} \quad (40)$$

$$\boxed{\bar{\sigma}_{vr} = -\frac{\mu}{2} S_{rr} \cos(2\phi)} \quad (41)$$

3.4 Layer-layer contact relationship

By Newton's third law, two surfaces in contact should experience equal and opposite traction forces. Because the surface normals are opposite, this means they should have equal stresses. It suffices to equate the normal pressures σ_{rr} because the transverse shear stresses are related above. Therefore, we get the following recursive relationships to define the boundary stresses, using piecewise gates based on layer contact boundaries:

$$\text{Outer surface: } \sigma_{rr}^+(u, v) = \begin{cases} \sigma_{rr}^-(u^+, v^+) & v \geq -\frac{W}{2} + 2\pi r \sin \phi \\ 0 & \text{otherwise} \end{cases} \quad (42)$$

$$\text{Inner surface: } \sigma_{rr}^-(u, v) = \begin{cases} \sigma_{rr}^+(u^-, v^-) & v \leq \frac{W}{2} - 2\pi r \sin \phi \\ 0 & \text{otherwise} \end{cases} \quad (43)$$

$$(44)$$

We can approximate this recursive relationship with a first order Taylor expansion. We can also rewrite the piecewise functions using Heaviside step functions $H(x)$, which is defined as $H(x) = 1$ for $x \geq 0$ and $H(x) = 0$ for $x < 0$. Let $H^+ := H(v + \frac{W}{2} - 2\pi r \sin \phi)$ and $H^- := H(v - \frac{W}{2} + 2\pi r \sin \phi)$ be the gates that define whether an element has an outer or inner neighbor, respectively. The outer layer contact relationship can be rewritten as:

$$\sigma_{rr}^-(u^+, v^+) = \sigma_{rr}^-(u + 2\pi r \cos \phi, v - 2\pi r \sin \phi) \quad (45)$$

$$\sigma_{rr}^+(u, v) \approx \left[\sigma_{rr}^-(u, v) + 2\pi r \cos \phi \frac{\partial \sigma_{rr}^-}{\partial u} - 2\pi r \sin \phi \frac{\partial \sigma_{rr}^-}{\partial v} \right] H^+ \quad (46)$$

Similarly, the inner layer contact relationship can be rewritten as:

$$\sigma_{rr}^-(u, v) \approx \left[\sigma_{rr}^+(u, v) - 2\pi r \cos \phi \frac{\partial \sigma_{rr}^+}{\partial u} + 2\pi r \sin \phi \frac{\partial \sigma_{rr}^+}{\partial v} \right] H^- \quad (47)$$

We then subtract Eq. 46 from Eq. 47 to obtain an expression for the difference in normal pressure across the thickness, $\Delta\sigma_{rr} = \sigma_{rr}^+ - \sigma_{rr}^-$. Depending on the values of H^+ and H^- (based on v , r ,

and ϕ), we have four cases:

$$\text{No contact} \quad \Delta\sigma_{rr} = 0 \quad (48)$$

$$\text{Outer contact only} \quad \Delta\sigma_{rr} = \sigma_{rr}^- + 2\pi r \cos \phi \frac{\partial \sigma_{rr}^-}{\partial u} - 2\pi r \sin \phi \frac{\partial \sigma_{rr}^-}{\partial v} \quad (49)$$

$$\text{Inner contact only} \quad \Delta\sigma_{rr} = -\sigma_{rr}^+ + 2\pi r \cos \phi \frac{\partial \sigma_{rr}^+}{\partial u} - 2\pi r \sin \phi \frac{\partial \sigma_{rr}^+}{\partial v} \quad (50)$$

$$\text{Both contacts} \quad \boxed{\Delta\sigma_{rr} = \pi r \cos \phi \frac{\partial S_{rr}}{\partial u} - \pi r \sin \phi \frac{\partial S_{rr}}{\partial v}} \quad (51)$$

It turns out that all cases are consistent with the “both contacts” case, Eq. 51, as long as the appropriate gates are applied to force σ_{rr}^+ and σ_{rr}^- to 0 individually when the corresponding layers are not in contact.

4 Compatibility

The previous sections have only dealt with kinematics and forces. Equilibrium only constrains that forces balance on every element of material; it says nothing about whether the resulting stress field is actually consistent with a single, continuous displacement of a real, unbroken sheet. A stress field can satisfy Eqs. (17)–(18) perfectly well and still correspond to a deformation that tears the ribbon or folds it through itself.

Let the thickness-averaged strain components as $\bar{\epsilon}_{uu}, \bar{\epsilon}_{vv}, \bar{\epsilon}_{uv}$. Because these come from differentiating a single, well-defined displacement field, they cannot be three independent functions; mixed partial derivatives of the (unknown, but assumed to exist) displacement field must commute:

$$\frac{\partial^2 \bar{\epsilon}_{uu}}{\partial v^2} + \frac{\partial^2 \bar{\epsilon}_{vv}}{\partial u^2} - 2 \frac{\partial^2 \bar{\epsilon}_{uv}}{\partial u \partial v} = 0. \quad (52)$$

Eq. (52) is a condition on *strain*, and our unknowns are *stress*. Hooke’s law, for a linear isotropic material under plane stress, is the bridge between the two. We introduce the material’s Young’s modulus E and Poisson ratio ν , and write the thickness-averaged Hooke’s law:

$$\bar{\epsilon}_{uu} = \frac{1}{E}(\bar{\sigma}_{uu} - \nu \bar{\sigma}_{vv}), \quad \bar{\epsilon}_{vv} = \frac{1}{E}(\bar{\sigma}_{vv} - \nu \bar{\sigma}_{uu}), \quad \bar{\epsilon}_{uv} = \frac{1+\nu}{E} \bar{\sigma}_{uv}. \quad (53)$$

Substituting Eq. (53) into Eq. (52) and multiplying through by E , the modulus cancels out completely. The resulting compatibility condition does not depend on how stiff the material is, only how it distributes strain between directions (ν):

$$\boxed{\frac{\partial^2 \bar{\sigma}_{uu}}{\partial v^2} - \nu \frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2} + \frac{\partial^2 \bar{\sigma}_{vv}}{\partial u^2} - \nu \frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2} - 2(1+\nu) \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v} = 0} \quad (54)$$

5 PDE formulation and closed-form solution

We now assemble the pieces above into a closed system for the stress field. The three equilibrium equations, Eqs. (17)–(19), are three PDEs for six unknown stress components at every point (u, v, ζ) . Section 3 uses Coulomb friction to reduce the problem by fixing σ_{ur}, σ_{vr} at the two surfaces $\zeta = \pm \frac{t}{2}$ directly in terms of σ_{rr} (Eqs. 36–39). By assuming the stress field varies linearly through the thickness, this entirely eliminates the unknowns σ_{ur}, σ_{vr} and reduces the remaining unknowns to functions of (u, v) alone. Although the in-plane stresses $\sigma_{uu}, \sigma_{uv}, \sigma_{vv}$ can be replaced by their thickness-averaged counterparts $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}$, the out-of-plane stress σ_{rr} affects the friction forces at each surface and cannot be averaged away. Instead, the two boundary values of σ_{rr} are treated as independent unknowns $\sigma_{rr}^+(u, v), \sigma_{rr}^-(u, v)$.

In total, we have 5 unknown stress components $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}, \sigma_{rr}^+, \sigma_{rr}^-$, all functions of (u, v) . We have 5 equations: the three in-plane equilibrium equations (Eqs. 17–18), the out-of-plane equilibrium equation (Eq. 19), the recursive layer-contact constraint on σ_{rr} (Eq. 51), and the compatibility condition (Eq. 54). Our boundary conditions are that the ribbon is traction-free at its edges $v = \pm W/2$ (i.e., $\sigma_{uv} = \sigma_{vv} = 0$), and that the ribbon has no transverse pressure at the outermost layer $u = L$ (i.e., $\sigma_{rr} = 0$), as well as the piecewise contact conditions on $\sigma_{rr}^\pm$ (Eqs. 42–43) as a function of v . This is a closed system of PDEs for the stress field, which we now solve in closed form.

Finally, we relate the components of the stress field to the macroscopic (Instron-exerted) force F , in order to arrive at the final force-displacement relationship.

5.1 In-plane thickness integration

We now integrate the three equilibrium equations through the thickness. Beginning with the $\hat{u}$ direction Eq (17),

$$\frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \left(\frac{\partial \sigma_{uu}}{\partial u} + \frac{\partial \sigma_{uv}}{\partial v} + \frac{\partial \sigma_{ur}}{\partial \zeta} + \kappa_{uu} \sigma_{ur} - \kappa_{uv} \sigma_{vr} \right) d\zeta = 0 \quad (55)$$

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} + \frac{1}{t} \sigma_{ur} \Big|_{-\frac{t}{2}}^{\frac{t}{2}} + \kappa_{uu} \bar{\sigma}_{ur} - \kappa_{uv} \bar{\sigma}_{vr} = 0 \quad (56)$$

Then substituting Eqs. (40)–(41) for the thickness-averaged transverse shear stresses, and expanding the trig identities in the curvature terms:

$$\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} - \frac{\mu \sin(2\phi)}{t} (\sigma_{rr}^+ - \sigma_{rr}^-) + \frac{\mu}{2} S_{rr} (\kappa_{uv} \cos(2\phi) - \kappa_{uu} \sin(2\phi)) = 0 \quad (57)$$

$$\boxed{\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} - \frac{\mu \sin(2\phi) \Delta \sigma_{rr}}{t} - \frac{\mu \sin(2\phi) S_{rr}}{4r} = 0} \quad (58)$$

Integrating the $\hat{v}$ direction Eq (18) gives a similar result:

$$\boxed{\frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} - \frac{\mu \cos(2\phi) \Delta \sigma_{rr}}{t} + \frac{\mu \cos(2\phi) S_{rr}}{4r} = 0} \quad (59)$$

These are two equations for the three unknowns $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}$ (given $\Delta\sigma_{rr}, S_{rr}$); Section 5.2 supplies the third.

5.2 Closing compatibility with friction: the Beltrami–Michell equation

Section 4 derived Eq. (54) without any reference to friction. We now combine the above transverse-shear-integrated in-plane balance equations to eliminate the mixed derivative $\frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v}$ from Eq. (54). Observe that Eqs. (58)–(59) already relate $\frac{\partial \bar{\sigma}_{uv}}{\partial v}$ and $\frac{\partial \bar{\sigma}_{uv}}{\partial u}$ to the other fields. We differentiate Eq. (58) with respect to u and Eq. (59) with respect to v , then add them together to isolate $2 \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v}$ on one side:

$$\begin{aligned} 2 \frac{\partial^2 \bar{\sigma}_{uv}}{\partial u \partial v} = & - \left(\frac{\partial^2 \bar{\sigma}_{uu}}{\partial u^2} + \frac{\partial^2 \bar{\sigma}_{vv}}{\partial v^2} \right) + \frac{\mu}{t} \left[\sin(2\phi) \frac{\partial \Delta\sigma_{rr}}{\partial u} + \cos(2\phi) \frac{\partial \Delta\sigma_{rr}}{\partial v} \right] \\ & + \frac{\mu}{4r} \left[\sin(2\phi) \frac{\partial S_{rr}}{\partial u} - \cos(2\phi) \frac{\partial S_{rr}}{\partial v} \right]. \end{aligned} \quad (60)$$

Substituting this sum into Eq. (54), the four second-derivative terms combine into a plain 2D Laplacian of $\bar{\sigma}_{uu} + \bar{\sigma}_{vv}$, giving the Beltrami–Michell equation

$$\boxed{\nabla^2(\bar{\sigma}_{uu} + \bar{\sigma}_{vv}) = (1 + \nu) \frac{\mu}{t} \left[\sin(2\phi) \frac{\partial \Delta\sigma_{rr}}{\partial u} + \cos(2\phi) \frac{\partial \Delta\sigma_{rr}}{\partial v} \right] + \frac{\mu}{4r} \left[\sin(2\phi) \frac{\partial S_{rr}}{\partial u} - \cos(2\phi) \frac{\partial S_{rr}}{\partial v} \right].} \quad (61)$$

Intuitively, this is a Poisson-type equation: the total in-plane stress $\bar{\sigma}_{uu} + \bar{\sigma}_{vv}$ can't vary however it likes—it has to curve in response to how unevenly the friction forcing is distributed, the way a temperature field curves in response to a heat source. Equations (58), (59), and (61) are now three equations for the three unknowns $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}$, closing the gap identified above.

5.3 Out-of-plane thickness integration

Next, we integrate the $\hat{r}$ direction Eq (19):

$$\frac{1}{t} \int_{-\frac{t}{2}}^{\frac{t}{2}} \left(\frac{\partial \sigma_{ur}}{\partial u} + \frac{\partial \sigma_{vr}}{\partial v} + \frac{\partial \sigma_{rr}}{\partial \zeta} - \kappa_{uu} \sigma_{uu} + 2\kappa_{uv} \sigma_{uv} - \kappa_{vv} \sigma_{vv} \right) d\zeta = 0 \quad (62)$$

$$\frac{\partial \bar{\sigma}_{ur}}{\partial u} + \frac{\partial \bar{\sigma}_{vr}}{\partial v} + \frac{1}{t} \Delta\sigma_{rr} - \kappa_{uu} \bar{\sigma}_{uu} + 2\kappa_{uv} \bar{\sigma}_{uv} - \kappa_{vv} \bar{\sigma}_{vv} = 0 \quad (63)$$

Substituting Eqs. (40)–(41) for $\bar{\sigma}_{ur}, \bar{\sigma}_{vr}$ gives,

$$\Delta\sigma_{rr} = t [\kappa_{uu} \sigma_{uu} - 2\kappa_{uv} \sigma_{uv} + \kappa_{vv} \sigma_{vv}] + \frac{t\mu}{2} \left[\sin(2\phi) \frac{\partial S_{rr}}{\partial u} + \cos(2\phi) \frac{\partial S_{rr}}{\partial v} \right] \quad (64)$$

This is one equation in the two unknowns $S_{rr}, \Delta\sigma_{rr}$ (equivalently, $\sigma_{rr}^+, \sigma_{rr}^-$); by itself it is not yet closed. We now bring in Eq (51) as an alternate expression for $\Delta\sigma_{rr}$ based on the recursive layer-to-layer contact relationship. Equating Eqs. (64) and (51) gives:

$$\left(\pi r \cos \phi - \frac{t\mu}{2} \sin(2\phi)\right) \frac{\partial S_{rr}}{\partial u} - \left(\pi r \sin \phi + \frac{t\mu}{2} \cos(2\phi)\right) \frac{\partial S_{rr}}{\partial v} = t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}] \quad (65)$$

The $t\mu$ corrections to the two coefficients are smaller than the leading $\pi r \cos \phi, \pi r \sin \phi$ terms by a factor of order t/r (comparing an explicit t against an explicit r), so at leading order they drop, leaving the closed transport equation for S_{rr} alone:

$$\boxed{\pi r \cos \phi \frac{\partial S_{rr}}{\partial u} - \pi r \sin \phi \frac{\partial S_{rr}}{\partial v} = t[\kappa_{uu}\sigma_{uu} - 2\kappa_{uv}\sigma_{uv} + \kappa_{vv}\sigma_{vv}]} \quad (66)$$

Given $\bar{\sigma}_{uu}, \bar{\sigma}_{uv}, \bar{\sigma}_{vv}$, Eq. (66) determines S_{rr} (a first-order transport equation, solvable along its characteristics using $\sigma_{rr}^+(L, v) = 0$), and $\Delta\sigma_{rr}$ then follows algebraically from Eq. (64) at the same leading order, $\Delta\sigma_{rr} \approx t[\kappa_{uu}\bar{\sigma}_{uu} - 2\kappa_{uv}\bar{\sigma}_{uv} + \kappa_{vv}\bar{\sigma}_{vv}]$. This is the self-contained 2-for-2 subsystem referred to above.

5.4 v-dependence reduction

Along the long edges of the ribbon, $v = \pm \frac{W}{2}$, the material is traction-free along the $\hat{v}$ face. This means we can enforce the following boundary conditions:

$$\bar{\sigma}_{uv}(u, \pm \frac{W}{2}) = 0, \quad \bar{\sigma}_{vv}(u, \pm \frac{W}{2}) = 0, \quad \bar{\sigma}_{vr}(u, \pm \frac{W}{2}) = 0. \quad (67)$$

In particular, σ_{vr} is directly related to S_{rr} (Eq. (41)), so this boundary condition implies that $\sigma_{rr}^\pm(u, \pm \frac{W}{2}) = 0$ individually¹ — and hence, since $\sigma_{rr}^\pm \leq 0$ always (Section ??), that $S_{rr}, \Delta\sigma_{rr}$ both vanish at $v = \pm \frac{W}{2}$ too, since a sum of two non-positive numbers is zero only if both are.

Rather than expand $S_{rr}, \Delta\sigma_{rr}$ directly, it is cleaner to first repackage the two combinations of them that actually appear in Eqs. (58)–(59):

$$P := \frac{\Delta\sigma_{rr}}{t} + \frac{S_{rr}}{4r}, \quad Q := \frac{\Delta\sigma_{rr}}{t} - \frac{S_{rr}}{4r}, \quad (68)$$

so that Eq. (58) reads $\frac{\partial \bar{\sigma}_{uu}}{\partial u} + \frac{\partial \bar{\sigma}_{uv}}{\partial v} = \mu \sin(2\phi)P$ and Eq. (59) reads $\frac{\partial \bar{\sigma}_{uv}}{\partial u} + \frac{\partial \bar{\sigma}_{vv}}{\partial v} = \mu \cos(2\phi)Q$. Since P, Q are each a linear combination of $\Delta\sigma_{rr}, S_{rr}$, both vanish at $v = \pm \frac{W}{2}$ too.

Take $\bar{\sigma}_{uu}, \bar{\sigma}_{vv}$ even in v (there is no reason for tension or normal stress to prefer one side of the centerline over the other). In Eq. (59), $\frac{\partial \bar{\sigma}_{vv}}{\partial v}$ is then odd; for $\frac{\partial \bar{\sigma}_{uv}}{\partial u}$ to balance it, $\bar{\sigma}_{uv}$ itself must be odd. Checking this against Eq. (58): with $\bar{\sigma}_{uv}$ odd, $\frac{\partial \bar{\sigma}_{uv}}{\partial v}$ is even, matching the even $\frac{\partial \bar{\sigma}_{uu}}{\partial u}$ — both equations balance in parity with no contradiction. (The free-edge condition alone does not decide this: both the even family $\cos\left(\frac{(2n-1)\pi v}{W}\right)$ and the odd family $\sin\left(\frac{2n\pi v}{W}\right)$ vanish at $v = \pm \frac{W}{2}$; it is the equilibrium equations, not the boundary condition, that fix which one $\bar{\sigma}_{uv}$ must use.) This forces P (which pairs with the even $\bar{\sigma}_{uu}$ -equation) to be even, and Q (which pairs with the odd

¹Technically, the relationship defined by Eq. (41) breaks when $\phi = 45^\circ$, but since every coefficient in the governing equations depends smoothly on ϕ with nothing singular at $\phi = 45^\circ$, the solution itself varies continuously through that point, so the value obtained in the limit $\phi \rightarrow 45^\circ$ from either side should apply there as well.

$\bar{\sigma}_{uv}$ -equation) to be odd. P, Q inherit the vanishing at $v = \pm \frac{W}{2}$ established above, so each lands in the edge-vanishing family matching its own parity.

Unlike $\bar{\sigma}_{uv}, \bar{\sigma}_{vv}, P, Q$, nothing forces $\bar{\sigma}_{uu}$ to vanish at the edges, so there is no boundary-condition argument pulling it toward $\cos(\pi v/W)$ specifically. Concretely: differentiating the leading odd mode of $\bar{\sigma}_{uv}$ produces $\cos(2\pi v/W)$ (frequency $\frac{2\pi}{W}$), not $\cos(\pi v/W)$ (frequency $\frac{\pi}{W}$, the frequency of P) — the two frequencies genuinely do not match, so no single-mode choice for $\bar{\sigma}_{uu}$ makes Eq. (58) balance term-by-term across all three sources exactly. Given that, the width-average (frequency 0, the lowest and simplest option, with no edge constraint to violate) is the most justifiable choice.

In summary, each of the five unknowns are reduced to a function of u alone:

$$\bar{\sigma}_{uu}(u, v) \approx \bar{\sigma}_{uu}(u), \quad \bar{\sigma}_{uv}(u, v) \approx a(u) \sin\left(\frac{2\pi v}{W}\right), \quad \bar{\sigma}_{vv}(u, v) \approx b(u) \cos\left(\frac{\pi v}{W}\right), \quad (69)$$

$$P(u, v) \approx p(u) \cos\left(\frac{\pi v}{W}\right), \quad Q(u, v) \approx q(u) \sin\left(\frac{2\pi v}{W}\right)$$

Then, substituting these expansions into Eqs. (58)–(59), (61), and (66) (the last two rewritten through P, Q , using $\Delta\sigma_{rr} = \frac{t}{2}(P + Q)$, $S_{rr} = 2r(P - Q)$) gives:

Eq. (58), in terms of P .

$$\boxed{\frac{d\bar{\sigma}_{uu}}{du} + \frac{2\pi}{W}a(u) \cos\left(\frac{2\pi v}{W}\right) - \mu \sin(2\phi) p(u) \cos\left(\frac{\pi v}{W}\right) = 0.} \quad (70)$$

Eq. (59), in terms of Q .

$$\boxed{\left[\frac{da}{du} - \mu \cos(2\phi) q(u)\right] \sin\left(\frac{2\pi v}{W}\right) - \frac{\pi}{W}b(u) \sin\left(\frac{\pi v}{W}\right) = 0.} \quad (71)$$

Eq. (61), in terms of P, Q . Using $\frac{\mu}{t} \sin(2\phi) \partial_u \Delta\sigma_{rr} + \frac{\mu}{4r} \sin(2\phi) \partial_u S_{rr} = \mu \sin(2\phi) \partial_u P$ and $\frac{\mu}{t} \cos(2\phi) \partial_v \Delta\sigma_{rr} - \frac{\mu}{4r} \cos(2\phi) \partial_v S_{rr} = \mu \cos(2\phi) \partial_v Q$:

$$\boxed{\frac{d^2\bar{\sigma}_{uu}}{du^2} + \left[\frac{d^2b}{du^2} - \left(\frac{\pi}{W}\right)^2 b(u)\right] \cos\left(\frac{\pi v}{W}\right) = (1 + \nu)\mu \sin(2\phi) \frac{dp}{du} \cos\left(\frac{\pi v}{W}\right) + (1 + \nu)\mu \cos(2\phi) \frac{2\pi}{W} q(u) \cos\left(\frac{2\pi v}{W}\right).} \quad (72)$$

Eq. (66), in terms of P, Q . Substituting $S_{rr} = 2r(P - Q)$, splitting by parity (even part gives an equation in P alone plus a Q -sourced correction; odd part gives Q alone plus a P -sourced correction):

$$(\text{even}) \quad 2\pi r^2 \cos \phi \frac{dp}{du} \cos\left(\frac{\pi v}{W}\right) - \frac{t}{2} p(u) \cos\left(\frac{\pi v}{W}\right) + \frac{4\pi^2 r^2 \sin \phi}{W} q(u) \cos\left(\frac{2\pi v}{W}\right) = 0, \quad (73)$$

$$(\text{odd}) \quad -2\pi r^2 \cos \phi \frac{dq}{du} \sin\left(\frac{2\pi v}{W}\right) - \frac{t}{2} q(u) \sin\left(\frac{2\pi v}{W}\right) + \frac{2\pi^2 r^2 \sin \phi}{W} p(u) \sin\left(\frac{\pi v}{W}\right) = 0. \quad (74)$$

5.5 Solving the reduced system

Collect the five amplitude functions from Eqs. (70), (71), (72), (73)–(74) into the first-order ODEs found by projecting each onto its own mode. Because Eq. (72) is second order in $b(u)$, introduce $\frac{db}{du}$ as an independent state so the whole system is first order. For convenience, let

$$\alpha := \frac{t}{4\pi r^2 \cos \phi}, \quad \beta := \frac{8 \sin \phi}{3W \cos \phi} \quad (75)$$

$$\gamma_p := \left[(1 + \nu) - \frac{8}{\pi^2} \right] \mu \sin(2\phi) \alpha \quad (76)$$

$$\gamma_q := - \left[(1 + \nu) - \frac{8}{\pi^2} \right] \mu \sin(2\phi) \beta + \frac{8(1 + \nu)\mu \cos(2\phi)}{3W} \quad (77)$$

for recurring combination of coefficients. Substituting $\frac{d^2 \bar{\sigma}_{uu}}{du^2} = \frac{2\mu \sin(2\phi)}{\pi} \frac{dp}{du}$ (from Eq. 70) and then $\frac{dp}{du} = \alpha p - \beta q$ into Eq. (72) gives the full system:

$$\left(\begin{array}{cc|cc|cc} \alpha & -\beta & 0 & 0 & 0 & 0 \\ \beta & -\alpha & 0 & 0 & 0 & 0 \\ \hline 0 & 0 & 0 & 1 & 0 & 0 \\ \gamma_p & \gamma_q & \left(\frac{\pi}{W}\right)^2 & 0 & 0 & 0 \\ \hline 0 & \mu \cos(2\phi) & \frac{8}{3W} & 0 & 0 & 0 \\ \hline \frac{2\mu \sin(2\phi)}{\pi} & 0 & 0 & 0 & 0 & 0 \end{array} \right) \begin{pmatrix} p \\ q \\ b \\ \frac{db}{du} \\ a \\ \bar{\sigma}_{uu} \end{pmatrix} = \frac{d}{du} \begin{pmatrix} p \\ q \\ b \\ \frac{db}{du} \\ a \\ \bar{\sigma}_{uu} \end{pmatrix} \quad (78)$$

The block structure (marked by the dividing lines) reveals that the matrix is *block lower-triangular*. p, q satisfy a closed 2×2 system with no dependence on anything below them; $b, \frac{db}{du}$ are driven only by p, q (plus their own block); a is driven only by b and q ; $\bar{\sigma}_{uu}$ is driven only by p . None of the later blocks ever feed back upward. This means Eq. (78) never has to be diagonalized as a single dense 6×6 problem — it solves in a cascade, each step using only the already-solved output of the previous one.

Step 1: the (p, q) block. This is exactly the self-contained pressure subsystem noted in Section 5's opening count. Its matrix $\begin{pmatrix} \alpha & -\beta \\ \beta & -\alpha \end{pmatrix}$ has trace zero and determinant $\beta^2 - \alpha^2$, giving eigenvalues

$$\lambda = \pm \sqrt{\alpha^2 - \beta^2}, \quad (79)$$

real (exponential growth/decay in u) if $|\alpha| > |\beta|$, imaginary (oscillatory in u) if $|\alpha| < |\beta|$. Either way, $p(u), q(u)$ come out as fixed linear combinations of $e^{\lambda u}$ and $e^{-\lambda u}$ (with coefficients set by the eigenvectors of that 2×2 matrix), with the two free constants fixed by boundary data on $\sigma_{rr}^\pm$ translated into P, Q at $u = L$ and $u = 0$, as in Section ??.

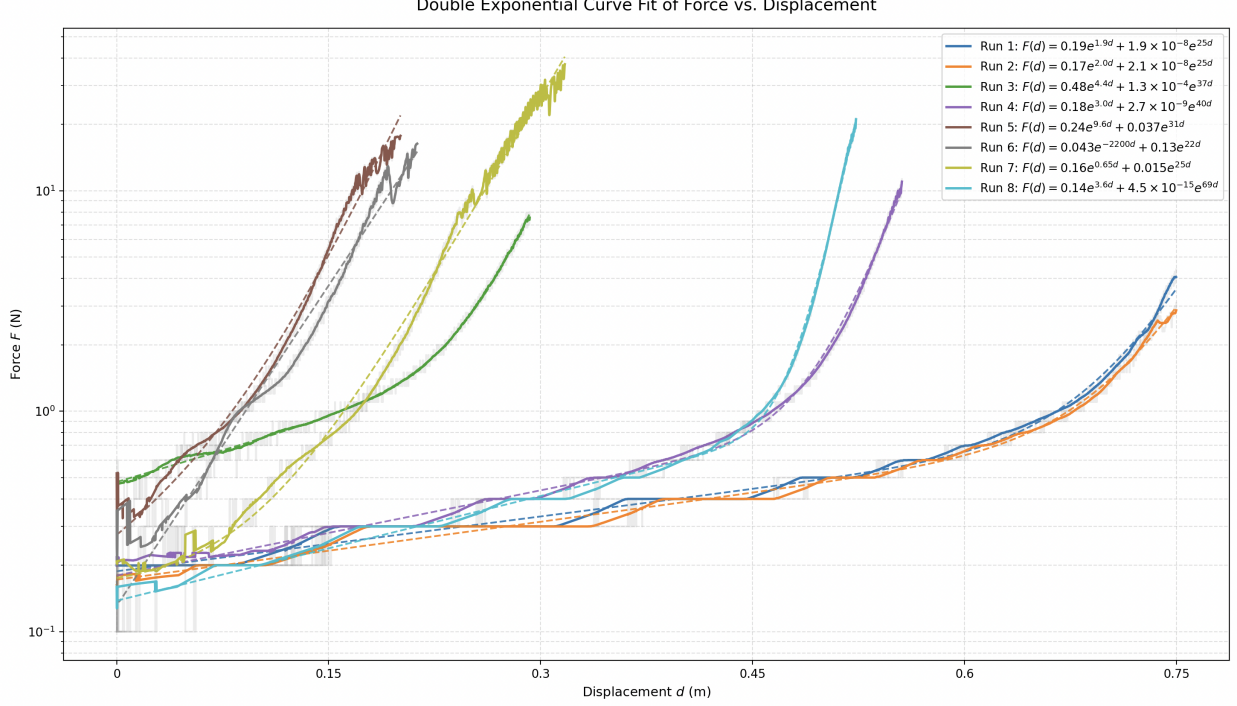


Figure 2: Instron force-displacement data, with raw quantized data (grey), smoothed data (solid colored), and curve fit (dashed colored).

Step 2: the $(b, \frac{db}{du})$ block. With $p(u), q(u)$ now known functions (sums of $e^{\pm\lambda u}$), this is a linear, constant-coefficient, second-order ODE for $b(u)$ with exponential forcing:

$$\frac{d^2b}{du^2} - \left(\frac{\pi}{W}\right)^2 b(u) = \gamma_p p(u) + \gamma_q q(u). \quad (80)$$

Its homogeneous solution is $b_h(u) = C_1 e^{\pi u/W} + C_2 e^{-\pi u/W}$; its particular solution is found by the standard undetermined-coefficients guess $b_p(u) \propto e^{\pm\lambda u}$ matching the forcing, unless $\lambda = \pm \frac{\pi}{W}$ exactly (resonance), in which case the usual $u e^{\pm\lambda u}$ correction applies.

Step 3: $a(u)$. With $b(u), q(u)$ known, $\frac{da}{du} = \mu \cos(2\phi) q(u) + \frac{8}{3W} b(u)$ is a direct integration.

Step 4: $\bar{\sigma}_{uu}(u)$. With $p(u)$ known, $\frac{d\bar{\sigma}_{uu}}{du} = \frac{2\mu \sin(2\phi)}{\pi} p(u)$ is likewise a direct integration.

6 Experimental validation

Will do some curve fitting and stuff once the final $F(x)$ form is derived. But for now here's a double exponential $F(x) = A \exp(-Bx) + C \exp(-Dx)$ fit to the Instron data, which fits remarkably well.